ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Oceans and Life** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** submarine relief	ocean floor has mountains, plains and trenches
2	**(d)** guyot	former seamount eroded by waves
3	**(b)** sea arch	arches form when sea caves are cut through a headland
4	**(d)** the tide	caused by gravitational pull
5	**(b)** Gulf of Kutch, Gulf of Khambhat and the Sundarbans	large tidal ranges
6	**(a)** cold current that supports icebergs near Canada	carries freezing water
7	**(c)** 169 names given by 13 member countries	e.g., Biparjoy and Remal
8	**(c)** Life Below Water	ocean protection
9	**(a)** 64 per cent	belong to all countries
10	**(c)** A is true but R is false	the warm Gulf Stream, not Labrador, warms Norway

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The landforms of the ocean floor (1); trenches and mid-ocean ridges (1).	1+1
12	It is shallow and rich in fish (1) and holds oil and gas (1).	1+1
13	A seamount is an underwater volcanic mountain with a peak (1); a guyot is a flat-topped seamount eroded by waves (1).	1+1
14	They erode headlands into caves, arches and stacks (1) and deposit sand to form beaches (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Continental shelf: shallow edge rich in fish and oil (1); mid-ocean ridge: huge underwater volcanic mountain chain (1); trench: deepest part of the ocean, e.g., Mariana (1).	1+1+1
16	Shallow shelves are rich fishing zones (1); deep trenches and abyssal plains are unsuitable for fishing but allow deep-water shipping routes (1); shallow banks and ridges are hazards that ships must chart (1).	1+1+1
17	Crest is the high peak and trough the low valley (1); wavelength is the distance between two crests and wave height the vertical distance from trough to crest (1); frequency is crests passing a point per second and period is the time for a crest to travel between two points (1).	1+1+1
	OR: Waves open cracks in a headland by hydraulic action (1); abrasion enlarges them into a cave which then breaks through to form an arch (1); the arch roof collapses leaving a stack which later wears down to a stump (1).	1+1+1
18	Waves deposit sand and pebbles to form beaches and bars (2); beaches are used for tourism, fishing activity and recreation (1).	2+1
	OR: It was caused by a massive undersea earthquake and struck many coasts without warning (1); about 2,30,000 lives were lost, including in Tamil Nadu, Andhra Pradesh, Keralam and the Andaman & Nicobar Islands (1); India set up the ITEWC with detection buoys and a real-time network (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Continental shelf, slope, abyssal plain, mid-ocean ridge, trench, seamount and guyot with brief features (2). (ii) Shelves are rich in life and resources; ridges and trenches create varied habitats (2). (iii) Mariana Trench (1).	2+2+1
	OR: (i) Shallow shelf supports fishing; deep sections guide shipping and cable routes; ridges and trenches are targets of exploration (3). (ii) Oil and gas, fish, sand and gravel (2).	3+2
20	(i) Alternate rise and fall of the ocean surface mostly caused by winds (1). (ii) Crack, sea cave, arch, stack and stump with the process of hydraulic action and abrasion (3). (iii) Beach (1).	1+3+1

	OR: (i) Sea stack (1). (ii) A sea arch's roof collapsed after waves eroded a headland (1). (iii) It will erode into a stump (1). (iv) Beach, formed by deposited sand and pebbles (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **Oceans and Life** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Mariana Trench	Pacific Ocean trench
2	**(c)** seamount	rises from the ocean floor
3	**(c)** crack → cave → arch → stack → stump	progressive erosion of a headland
4	**(d)** the Sun, Earth and Moon are in a straight line	full moon and new moon
5	**(d)** ensure enough water depth and avoid running aground	tidal information is vital for navigation
6	**(b)** warm and cold currents meet	nutrient-rich waters
7	**(b)** living and non-living resources	fish vs oil, gas and minerals
8	**(c)** 1,852 metres	about one minute of latitude
9	**(c)** innocent passage	must not threaten peace or security
10	**(a)** Both A and R are true and R is the correct explanation of A	combined pull gives higher tides

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wavelength is the horizontal distance between two consecutive crests (1); wave period is the time taken by a crest to travel between two points (1).	1+1
12	A giant wave caused mainly by an undersea earthquake (1); the maximum vertical height it reaches above sea level on the shore (1).	1+1
13	High tide is the highest water level covering much of the shore (1); low tide is the lowest level when water moves away from the shore (1).	1+1
14	On full moon and new moon days when the Sun, Moon and Earth are aligned (1); the combined gravitational pull is strongest (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Spring tides occur when the Sun, Earth and Moon are in a line on full and new moon days (1); neap tides occur at first and third quarters when pulls act in different directions (1); spring tides have a larger range than neap tides (1).	1+1+1
16	They allow ships to enter harbours safely (1); they bring fish closer to the shore, improving catches (1); the rise and fall of water can generate tidal electricity (1).	1+1+1
17	To know the water depth and tidal range at the port (1); to avoid grounding during low tide (1); to plan safe docking and departure times (1).	1+1+1
	OR: Winds blowing over the surface (1); Earth's rotation through the Coriolis force (1); differences in temperature and salinity, and the shape of continents (1).	1+1+1
18	Warm currents flow from warmer regions and are shown in red; example Gulf Stream (1.5); cold currents flow from colder regions and are shown in blue; example Labrador Current (1.5).	1½+1½
	OR: Warm currents raise temperatures of nearby coasts and keep ports ice-free (1.5); cold currents lower temperature and may carry icebergs, while mixing of currents creates fishing grounds (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Rhythmic rise and fall of the sea, twice a day in most places (1). (ii) Spring tides when Sun, Earth and Moon align on full and new moon with combined pull; neap tides at quarter moons with pulls at different angles (3). (iii) Depth and tidal range decide safe berthing (1).	1+3+1
	OR: (i) Navigation, fishing, tidal power and access to places like the Dwarka temple at low tide (3). (ii) Gulf of Kutch and Gulf of Khambhat (2).	3+2

20	(i) The channel is shallow (1). (ii) High tide (1). (iii) About 12 m (1). (iv) Strong tidal currents of 5–6 knots can endanger boats and cause grounding (2).	1+1+1+2
	OR: (i) Continuous streams of water flowing in definite directions (1). (ii) Winds, Coriolis force, temperature and salinity differences and continent shape (2). (iii) Warm from warmer regions (red); cold from colder regions (blue) (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND